Chapter 8 : First-order Logic

AI 510 - 2025

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Outline

Why First Order Logic?

Syntax and semantics of FOL

Fun with sentences

Wumpus world in FOL

Propositional logic: pros and cons

PROS:

- Declarative: knowledge and inference are separate, and inference is entirely domain independent

- Allows partial/disjunctive/negated information (unlike most data structures and databases)

- Compositional
meaning of $A \wedge B$ is derived from meaning of A and of B

- Context-independent

CONS:

- Limited expressive power

e.g. $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

First order logic

In propositional logic world contains facts,
in first-order logic the world contains:

- Objects: people, houses, numbers ...

- Relations: (unary) red, round, prime . . .
  (binary) brother of, bigger than, occurred after...

- Functions: father of, owned by, one more than ...

Difference betweeen relations and functions?

- “One plus two equals three.”

  Objects: one, two, three, one plus two;
  Relation: equals;
  Function: plus.

- “Squares neighboring the wumpus are smelly.”

  Objects: wumpus, squares;
  Property: smelly;
  Relation: neighboring.

- “Evil King John ruled England in 1200.”

  Objects: John, England, 1200;
  Relation: ruled during;
  Properties: evil, king.

Logics in general

Language	Ontological Commitment	Epistemological Commitment
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief
Fuzzy logic	facts + degree of truth	known interval value

what it assumes about the nature of reality

the possible states of knowledge that it allows with respect to each fact

FOL Syntax

$Sentence \rightarrow Atomic\ Sentence \mid Complex\ Sentence$

$Atomic\ Sentence \rightarrow Predicate \mid Predicate(Term,\ldots) \mid Term = Term$

$$
\begin{aligned}
Complex\ Sentence \rightarrow\ & (\ Sentence\ ) \\
\mid\ & \neg\ Sentence \\
\mid\ & Sentence \wedge Sentence \\
\mid\ & Sentence \vee Sentence \\
\mid\ & Sentence \Rightarrow Sentence \\
\mid\ & Sentence \Leftrightarrow Sentence \\
\mid\ & Quantifier\ Variable, \ldots\ Sentence
\end{aligned}
$$

$$
\begin{aligned}
Term \rightarrow\ & Function(Term,\ldots) \\
\mid\ & Constant \\
\mid\ & Variable
\end{aligned}
$$

$Quantifier \rightarrow \forall \mid \exists$

$Constant \rightarrow A \mid X_1 \mid John \mid \cdots$

$Variable \rightarrow a \mid x \mid s \mid \cdots$

$Predicate \rightarrow True \mid False \mid After \mid Loves \mid Raining \mid \cdots$

$Function \rightarrow Mother \mid LeftLeg \mid \cdots$

Operator Precedence : $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

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Syntax of FOL: Basic elements

Constants	KingJohn, 2 ...
Predicates	Brother, >, ...
Functions	Sqrt, LeftLegOf, ...
Variables	x, y, a, b, ...
Connectives	$\wedge \quad \vee \quad \neg \quad \Rightarrow \quad \Leftrightarrow$
Equality	$=$
Quantifiers	$\forall \; \exists$

Predicates ... Relations

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Terms - objects

Refer to **objects :**

- **constant :** Name of the specific object ex. King John

- **function(term,…) :** Specify the object using an "expression"
  ex. King John's left leg, that is , LeftLegOf(King John)

- **variable** : Gerneric object

Atomic sentences – atomic facts

state **facts**:

- predicate , zero-ary predicates : like propositional variables
- predicate ($term_1$,...,$term_n$), n-ary predicates
- $term_1$ = $term_2$

E.g., Brother (KingJohn,RichardTheLionheart)

> (Length(LeftLegOf(Richard)),Length(LeftLegOf(KingJohn)))

2+2=4

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Complex sentences – complex facts

Complex sentences are made from atomic sentences using connectives

$\neg S, \quad S_1 \wedge S_2, \quad S_1 \vee S_2, \quad S_1 \Rightarrow S_2, \quad S_1 \Leftrightarrow S_2$

E.g. $Sibling(KingJohn, Richard) \Rightarrow Sibling(Richard, KingJohn)$
$>(1,2) \vee \leq(1,2)$
$>(1,2) \wedge \neg >(1,2)$

Quantifiers

"For all" – **Universal quantifier**

$\forall\ x\ King(x) \Rightarrow Person(x)$

The sentence $\forall\ x\ P$

states that

P is true for every object x.

"there exists" – **Existential quantifier**

$\exists\ x\ Crown(x) \land OnHead(x, John)$

$\exists x\ P$

states that

P is true for at least one object x.

Fun with sentences

Brothers are siblings

Fun with sentences

Brothers are siblings

$\forall x, y \;\; Brother(x, y) \;\; \Rightarrow \;\; Sibling(x, y).$

"Sibling" is symmetric

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"Sibling" is symmetric

$\forall x, y \;\; Sibling(x, y) \;\; \Leftrightarrow \;\; Sibling(y, x).$

One's mother is one's female parent

Fun with sentences

Brothers are siblings

$\forall x, y \quad Brother(x, y) \quad \Rightarrow \quad Sibling(x, y).$

"Sibling" is symmetric

$\forall x, y \quad Sibling(x, y) \quad \Leftrightarrow \quad Sibling(y, x).$

One's mother is one's female parent

$\forall x, y \quad Mother(x, y) \quad \Leftrightarrow \quad (Female(x) \land Parent(x, y)).$

A first cousin is a child of a parent's sibling

Fun with sentences

Brothers are siblings

$\forall x,y \quad Brother(x,y) \quad \Rightarrow \quad Sibling(x,y).$

"Sibling" is symmetric

$\forall x,y \quad Sibling(x,y) \quad \Leftrightarrow \quad Sibling(y,x).$

One's mother is one's female parent

$\forall x,y \quad Mother(x,y) \quad \Leftrightarrow \quad (Female(x) \land Parent(x,y))$

A first cousin is a child of a parent's sibling

$\forall x,y \quad FirstCousin(x,y) \quad \Leftrightarrow \quad \exists\, p,ps \quad Parent(p,x) \land Sibling(ps,p) \land Parent(ps,y)$

Truth in first-order logic

Sentences are true with respect to a **model**

Set of Objects
(domain)
+
Set Relations
+
interpretation

A function that associates

constant symbols → objects

predicate symbols → relations

function symbols → functional relations

An atomic sentence predicate($term_1$,...,$term_n$) is true iff the objects referred to by $term_1$,...,term are in the relation referred to by predicate

Truth example

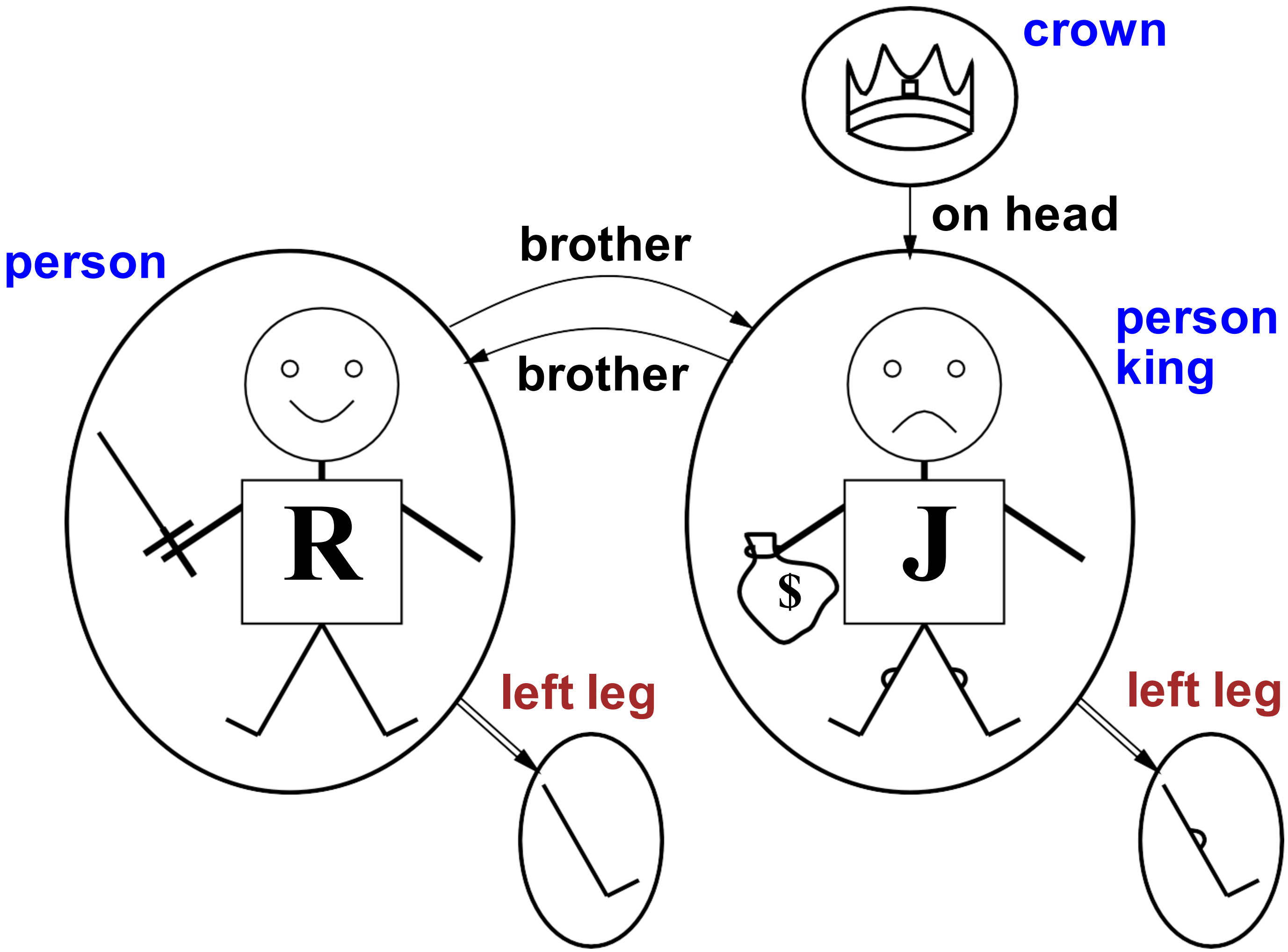
Given the model **M** , such that :
Domain(M) = Richard the Lionheart , the evil King Jhon
Relations(M) = the brotherhood relation

Consider the interpretation **i** in which :

Richard → Richard the Lionheart
John → the evil King John
Brother → the brotherhood relation

Under this interpretation, Brother(Richard,John) is true just in case Richard the Lionheart and the evil King John are in the brotherhood relation in **M**

Models for FOL: Example



A model containing five objects, two binary relations (brother and on-head), three unary relations (person, king, and crown), and one unary function (left-leg).

Models for FOL: Lots!

Propositional logic : entailment computed by enumerating models
(model checking)

FOL: we can try …

For each number of domain elements n from 1 to ∞

For each k-ary predicate P_k in the vocabulary For

each possible k-ary relation on n objects

For each constant symbol C in the vocabulary

For each choice of referent for C from n objects …

Not easy!

Truth of the Universal quantification

$\forall$ variables.sentences

$\forall x.\ P$ is true in a model **M** iff P is true with x being each possible object in the model.

EX. Everyone at Berkeley is smart:

$\forall x \quad At(x, Berkeley) \Rightarrow Smart(x)$

Roughly speaking, equivalent to the conjunction of instantiations of P

$$
\begin{aligned}
& (At(KingJohn, Berkeley) \Rightarrow Smart(KingJohn)) \\
\wedge\ & (At(Richard, Berkeley) \Rightarrow Smart(Richard)) \\
\wedge\ & (At(Berkeley, Berkeley) \Rightarrow Smart(Berkeley)) \\
\wedge\ & \ldots
\end{aligned}
$$

A common mistake to avoid

Typically, $\Rightarrow$ is the main connective with $\forall$

Common mistake: using $\wedge$ as the main connective with $\forall$:

$$\forall x.\ At(x, Berkeley) \wedge Smart(x)$$

means "Everyone is at Berkeley and everyone is smart"

Truth of the Existential quantification

$\exists$ *.variables.sentences*

$\exists$ x.P is true in a model m iff P is true with x being some possible object in the model

EX. Someone at Stanford is smart:

$\exists x \quad At(x, Stanford) \land Smart(x)$

Roughly speaking, equivalent to the disjunction of instantiations of *P*

$$
\begin{aligned}
& (At(KingJohn, Stanford) \land Smart(KingJohn)) \\
\lor \; & (At(Richard, Stanford) \land Smart(Richard)) \\
\lor \; & (At(Stanford, Stanford) \land Smart(Stanford)) \\
\lor \; & \ldots
\end{aligned}
$$

Another common mistake to avoid

Typically, $\wedge$ is the main connective with $\exists$

Common mistake: using $\Rightarrow$ as the main connective with $\exists$:

$$\exists x \quad At(x, Stanford) \Rightarrow Smart(x)$$

is true if there is anyone who is not at Stanford!

Properties of quantifiers

$\forall x \forall y$ is the same as $\forall y \forall x$

$\exists x \exists y$ is the same as $\exists x \exists y$

$\exists x \forall y$ is NOT the same as $\forall y \exists x$

EX.

$\exists x \quad \forall y \quad Loves(x, y)$

"There is a person who loves everyone in the world"

$\forall y \quad \exists x \quad Loves(x, y)$

"Everyone in the world is loved by at least one person"

Quantifier duality: each can be expressed using the other

$\forall x.Likes(x, IceCream)$ $\quad\quad$ $\neg \exists x.\neg Likes(x, IceCream)$

$\exists x.Likes(x, Broccoli)$ $\quad\quad$ $\neg \forall x.\neg Likes(x, Broccoli)$

Truth of Equality

$term_1 = term_2$ is true under a given interpretation
if and only if $term_1$ and $term_2$ refer to the same object

E.g., $1 = 2$ and $\forall x.\ \times(Sqrt(x), Sqrt(x)) = x$ are satisfiable
$2 = 2$ is valid

E.g., definition of (full) *Sibling* in terms of *Parent*:

$$\forall x,y \quad Sibling(x,y) \quad \Leftrightarrow$$

$$[\neg(x = y) \land \exists m,f \;\; \neg(m = f) \land Parent(m,x) \land Parent(f,x) \land Parent(m,y) \land Parent(f,y)]$$

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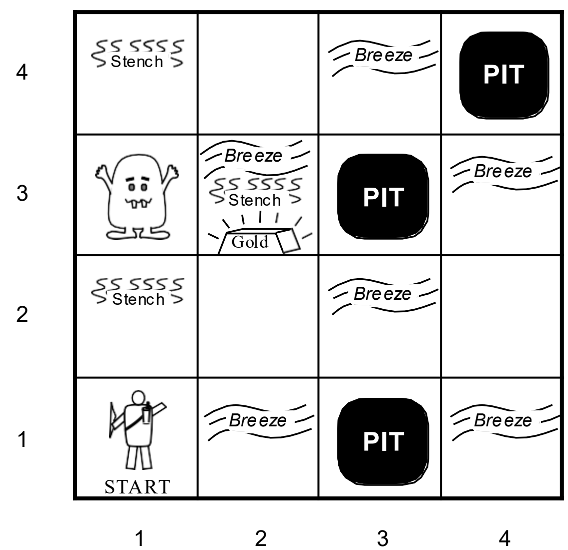
The Wumpus World

The wumpus world is a cave consisting of rooms connected by passageways. Lurking somewhere in the cave is the ***terrible wumpus****, a beast that eats anyone who enters its room.*
Some rooms contain bottomless ***pits*** *that will trap anyone who wanders into these rooms (except for the wumpus, which is too big to fall in).*
The only redeeming feature of this bleak environment is the possibility of finding a heap of ***gold****.*



The Wumpus World

The wumpus agent receives a **percept vector** with 5 elements, at a certain time.

Percept([Stench,Breeze,Glitter,None,None],5)

Binary Predicate

Constants in a list

integers as constants

The can be represented by logical terms:

Turn(Right), Turn(Left), Forward, Shoot, Grab, Climb.

Interacting with FOL KBs

Suppose a wumpus-world agent is using a FOL KB
and perceives a smell and a breeze (but no glitter) at $t = 5$:

Tell(KB, Percept([Smell, Breeze, None], 5))
Ask(KB, $\exists$ a.Action(a,5))
Answer: Yes,
{a/Shoot} ← substitution (binding list)

Given a sentence S and a substitution σ,
$S\sigma$ denotes the result of plugging σ into S; e.g.,
S = Smarter(x, y)
σ = {x/Hillary, y/Bill}
$S\sigma$ = Smarter(Hillary, Bill)

Ask(KB, S) returns some/all σ such that $KB \models S\sigma$

Knowledge base for the wumpus world

"Perception"

$\forall$ t, s,g,w,c. Percept([s,Breeze,g,w,c], t) $\Rightarrow$ Breeze(t)

$\forall$t, s,g,w,c. Percept([s,None,g,w,c], t) $\Rightarrow \neg$ Breeze(t)

$\forall$t, s,b,w,c . Percept([s,b,Glitter,w,c], t) $\Rightarrow$ Glitter(t)

$\forall$t, s,b,w,c . Percept([s,b,None,w,c], t) $\Rightarrow$ $\neg$ Glitter(t)

"Reflex" behaviour

$$\forall t.\ AtGold(t) \Rightarrow Action(Grab, t)$$

Representing the Wumpus World in FOL

Squares :
complex term in which the row and column appear as integers
e.g. [1,2]

Adjacent relation:

$$\forall x,y,a,b\ Adjacent([x,y],[a,b]) \Leftrightarrow$$
$$(x = a \land (y = b-1 \lor y = b+1)) \lor (y = b \land (x = a-1 \lor x = a+1))$$

We also need…

A <u>predicate</u> **Pit** that is true of squares containing pits.

Since there is exactly one wumpus, a <u>constant</u> **Wumpus.**

$At(Agent, s, t)$: agent is at square s at time t.

$\forall t\ At(Wumpus, [1,3], t)$: fix the wumpus to a specific location.

$\forall x, s1, s2, t\ At(x, s1, t) \land At(x, s2, t) \Rightarrow s1 = s2$:
States that objects can be at only one location at a time.

$\forall s, t\ At(Agent, s, t) \land Breeze(t) \Rightarrow Breezy(s)$:
if the agent is at a square and perceives a breeze, then that square is breezy.

$\forall s\ Breezy(s) \Leftrightarrow \exists r\ Adjacent(r, s) \land Pit(r)$:
If a square is breezy then there is a pit in one of the adjecent squares.

Summary

First-order logic:

– objects and relations are semantic primitives

– syntax: constants, functions, predicates, equality, quantifiers

Increased expressive power: sufficient to define

wumpus world

Read 8.1, 8.2, 8.3.4 of Artifitial Intelligence - A modern approach

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